Printed Pages: 02 **Sub Code: RPH-734**

Paper Id: 231874 **Roll No.** | | | | | | | | | | |

B. PHARM
(SEM VII) THEORY EXAMINATION 2022-23
PHARMACEUTICS-IX (BIOPHARMACEUTICS & PHARMACOKINETICS)

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections.

SECTION A

1. Attempt *all* questions in brief. **2 x 7 = 14**

(a) Define drug absorption and distribution coefficient.

(b) Write full name IVIVC and its significance.

(c) Give definition of volume of distribution and drug distribution.

(d) Define biopharmaceutics and pharmacokinetic

(e) Write Difference between intra venous (I.V.) bolus and I.V. infusion.

(f) Differentiate Zero order and first order absorption rate.

(g) What is compartment model, classify various type of pharmacokinetics model.

SECTION B

2. Attempt any *three* of the following: **7 x 3 = 21**

(a) Write detail note on regulatory requirements for conduction of bioequivalence studies.

(b) Write a detail note on in-vivo in-vitro correlation (IVIVC) and its significance.

(c) Define drug distribution? Explain physicochemical and pharmaceutical factors which influencing drug distribution.

(d) What is bio-equivalence. Write the various methods involved in the determination of bio-equivalence.

(e) Find out the absorption rate constant following two compartment model by Loo Riegelmann method give the advantages of this method.

SECTION C

3. Attempt any *one* part of the following: **7 x 1 = 7**

(a) Describe biopharmaceutics and pharmacokinetics and their role in formulation development.

(b) Calculate the various pharmacokinetic parameters and absorption rate constant with the help of urine data method.

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4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write a detail note compartment and non compartment model and its significance.

(b) Write significance of plasma drug concentration in calculation of various pharmacokinetic parameters.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Explain different mechanism of drug absorption.

(b) Discuss i*n-vivo in-vitro* correlation (IVIVC) and its significance.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Describe pharmacokinetics of drug absorption by zero order absorption rate constant.

(b) Write physicochemical and pharmaceutical factor which increase bioavailability of drug and give any example.

7. Attempt any *one* part of the fpoi8ollowing: 7 x 1 = 7

(a) Write a note on pharmacokinetic parameters from urine data after drug administration by intra venous (I.V.) bolus.

(b) Discuss about several methods of dosage adjustment in patient with renal disease.

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